CUSTOMER SEGMENTATION & MARKETING ANALYTICS

Data Storytelling, Dashboarding & Business Insights

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Understanding Customer Behavior at Scale

This project analyzes customer demographics, spending patterns, purchase channels, campaign responses, and satisfaction signals to support data-driven decision-making across marketing and sales teams.

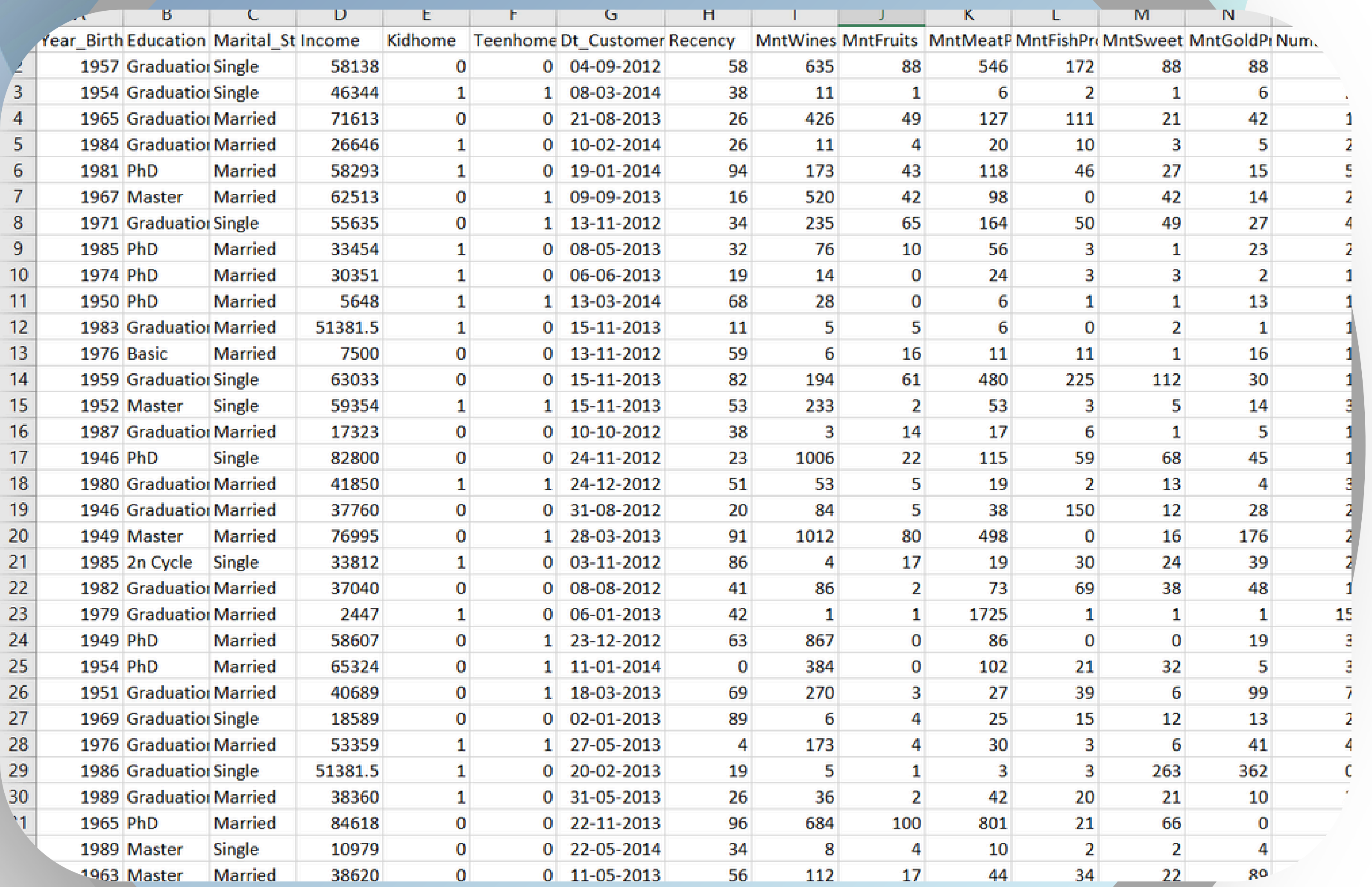
Behavioral Analysis

Campaign Effectiveness

Strategic Segmentation

Dataset Overview

The dataset combines demographic, transactional, and behavioral data across five key dimentions.



	Year_Birth	Education	Marital_St	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	MntFruits	MntMeatP	MntFishPr	MntSweet	MntGoldPr
2	1957	Graduatioı	Single	58138	0	0	04-09-2012	58	635	88	546	172	88	88
3	1954	Graduatioı	Single	46344	1	1	08-03-2014	38	11	1	6	2	1	6
4	1965	Graduatioı	Married	71613	0	0	21-08-2013	26	426	49	127	111	21	42
5	1984	Graduatioı	Married	26646	1	0	10-02-2014	26	11	4	20	10	3	5
6	1981	PhD	Married	58293	1	0	19-01-2014	94	173	43	118	46	27	15
7	1967	Master	Married	62513	0	1	09-09-2013	16	520	42	98	0	42	14
8	1971	Graduatioı	Single	55635	0	1	13-11-2012	34	235	65	164	50	49	27
9	1985	PhD	Married	33454	1	0	08-05-2013	32	76	10	56	3	1	23
10	1974	PhD	Married	30351	1	0	06-06-2013	19	14	0	24	3	3	2
11	1950	PhD	Married	5648	1	1	13-03-2014	68	28	0	6	1	1	13
12	1983	Graduatioı	Married	51381.5	1	0	15-11-2013	11	5	5	6	0	2	1
13	1976	Basic	Married	7500	0	0	13-11-2012	59	6	16	11	11	1	16
14	1959	Graduatioı	Single	63033	0	0	15-11-2013	82	194	61	480	225	112	30
15	1952	Master	Single	59354	1	1	15-11-2013	53	233	2	53	3	5	14
16	1987	Graduatioı	Married	17323	0	0	10-10-2012	38	3	14	17	6	1	5
17	1946	PhD	Single	82800	0	0	24-11-2012	23	1006	22	115	59	68	45
18	1980	Graduatioı	Married	41850	1	1	24-12-2012	51	53	5	19	2	13	4
19	1946	Graduatioı	Married	37760	0	0	31-08-2012	20	84	5	38	150	12	28
20	1949	Master	Married	76995	0	1	28-03-2013	91	1012	80	498	0	16	176
21	1985	2n Cycle	Single	33812	1	0	03-11-2012	86	4	17	19	30	24	39
22	1982	Graduatioı	Married	37040	0	0	08-08-2012	41	86	2	73	69	38	48
23	1979	Graduatioı	Married	2447	1	0	06-01-2013	42	1	1	1725	1	1	1
24	1949	PhD	Married	58607	0	1	23-12-2012	63	867	0	86	0	0	19
25	1954	PhD	Married	65324	0	1	11-01-2014	0	384	0	102	21	32	5
26	1951	Graduatioı	Married	40689	0	1	18-03-2013	69	270	3	27	39	6	99
27	1969	Graduatioı	Single	18589	0	0	02-01-2013	89	6	4	25	15	12	13
28	1976	Graduatioı	Married	53359	1	1	27-05-2013	4	173	4	30	3	6	41
29	1986	Graduatioı	Single	51381.5	1	0	20-02-2013	19	5	1	3	3	263	362
30	1989	Graduatioı	Married	38360	1	0	31-05-2013	26	36	2	42	20	21	10
31	1965	PhD	Married	84618	0	0	22-11-2013	96	684	100	801	21	66	0
	1989	Master	Single	10979	0	0	22-05-2014	34	8	4	10	2	2	4
	1963	Master	Married	38620	0	0	11-05-2013	56	112	17	44	34	22	89

Demographics Age, education, marital status, and household composition

Income & Expenses Annual income and multi-category expenditure data

Responses Customer service complaint frequency and patterns

Complaints Acceptance rates across marketing campaigns

Purchase Channels Web, catalog, and in-store transaction records

KPI Overview

663 active customers in dataset

52K Annual household income

Average Income	Response Rate	Average Spending	Total Customers	Complaint Rate
52,238K	50.38%	605.8	663	0.9375%

50% Campaign Acceptance

605.8 Average Spending

0.9% Customers with complaints

KPI Calculations

- Total Customers = COUNT(Customer_ID)
- Average Income = AVG(Income)
- Average Spending = AVG(Total_Spending)
- Response Rate = SUM(Response) / Total Customers *100
- Complaint Rate = SUM(Complain) / Total Customers* 100

```
# KPI 1: Average Customer Spending
total_spending = df['Total_Spending'].sum()
total_customers = len(df)
avg_customer_spending = total_spending / total_customers
print("Average Customer Spending:", avg_customer_spending)
# KPI 2: Campaign Response Rate
responded_customers = df['Response'].sum()
response_rate = (responded_customers / total_customers) * 100
print("Campaign Response Rate:", response_rate, "%")
# KPI 3: Average Income
avg_income = df['Income'].mean()
print("Average Income:", avg_income)
# KPI 4: Web Purchase Rate
total_web = df['NumWebPurchases'].sum()
total_store = df['NumStorePurchases'].sum()
```

Customer Segment Distribution

Customers were segmented based on spending behavior into High Value, Medium Value, and Low Value groups.

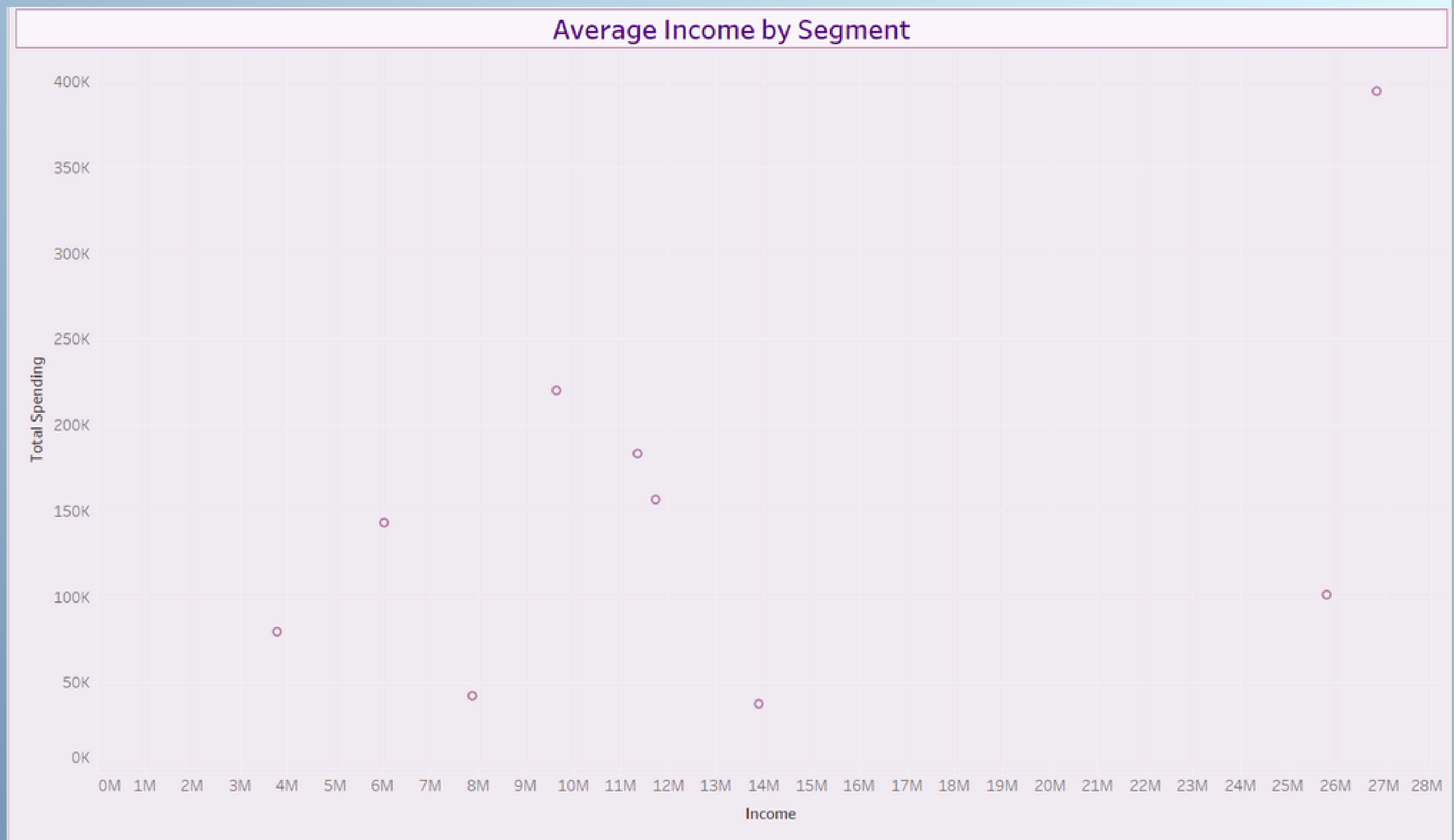
Key Findings:

- Low Value customers represent the largest customer segment.
- Medium Value customers form the second largest group.
- High Value customers represent the smallest segment.

Business Insight:

Although High Value customers are fewer in number, they contribute significantly to revenue generation. Low and Medium Value customers represent opportunities for targeted upselling and retention campaigns.

Average Income by Segment



A scatter plot was used to examine how customer income influences total spending.

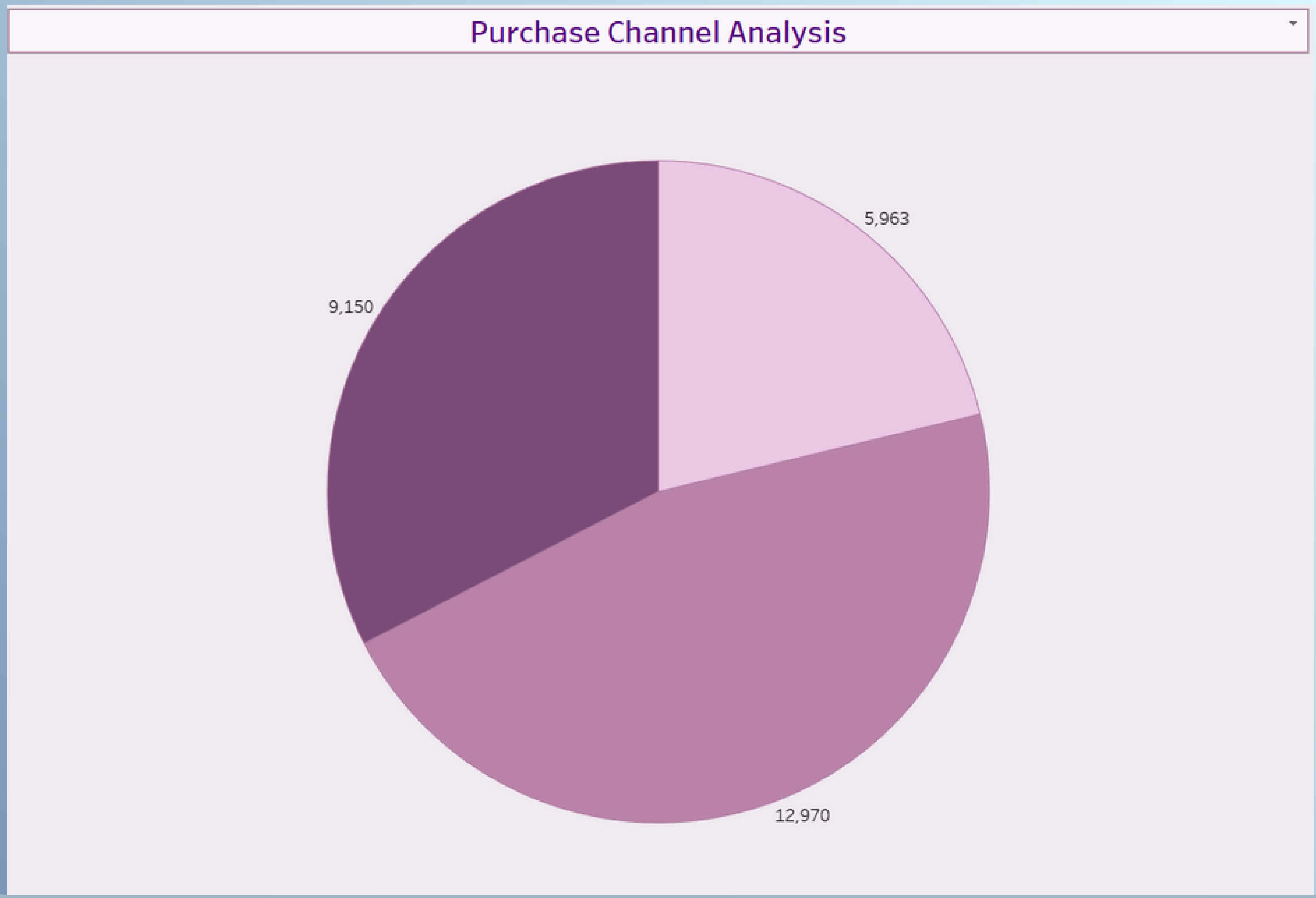
Key Findings:

- Customers with higher incomes generally exhibit higher spending behavior.
- A positive relationship exists between income and total spending.
- Spending patterns vary across customer groups, indicating diverse purchasing behaviors.

Business Insight:

Income is an important factor influencing customer purchasing power. Businesses can use income-based segmentation to create more effective marketing strategies and personalized offers.

Purchase Channel Analysis



Customer purchasing activity was analyzed across Web, Store, and Catalog channels.

Key Findings:

- Store purchases account for the largest share of transactions.
- Web purchases contribute significantly to customer activity.
- Catalog purchases represent the smallest purchase channel.

Business Insight:

Physical stores remain the dominant sales channel, while web purchases demonstrate strong digital engagement. Continued investment in both channels can help maximize customer reach and revenue.

Customer Segment Response

Campaign responses were evaluated across customer segments to understand customer engagement.

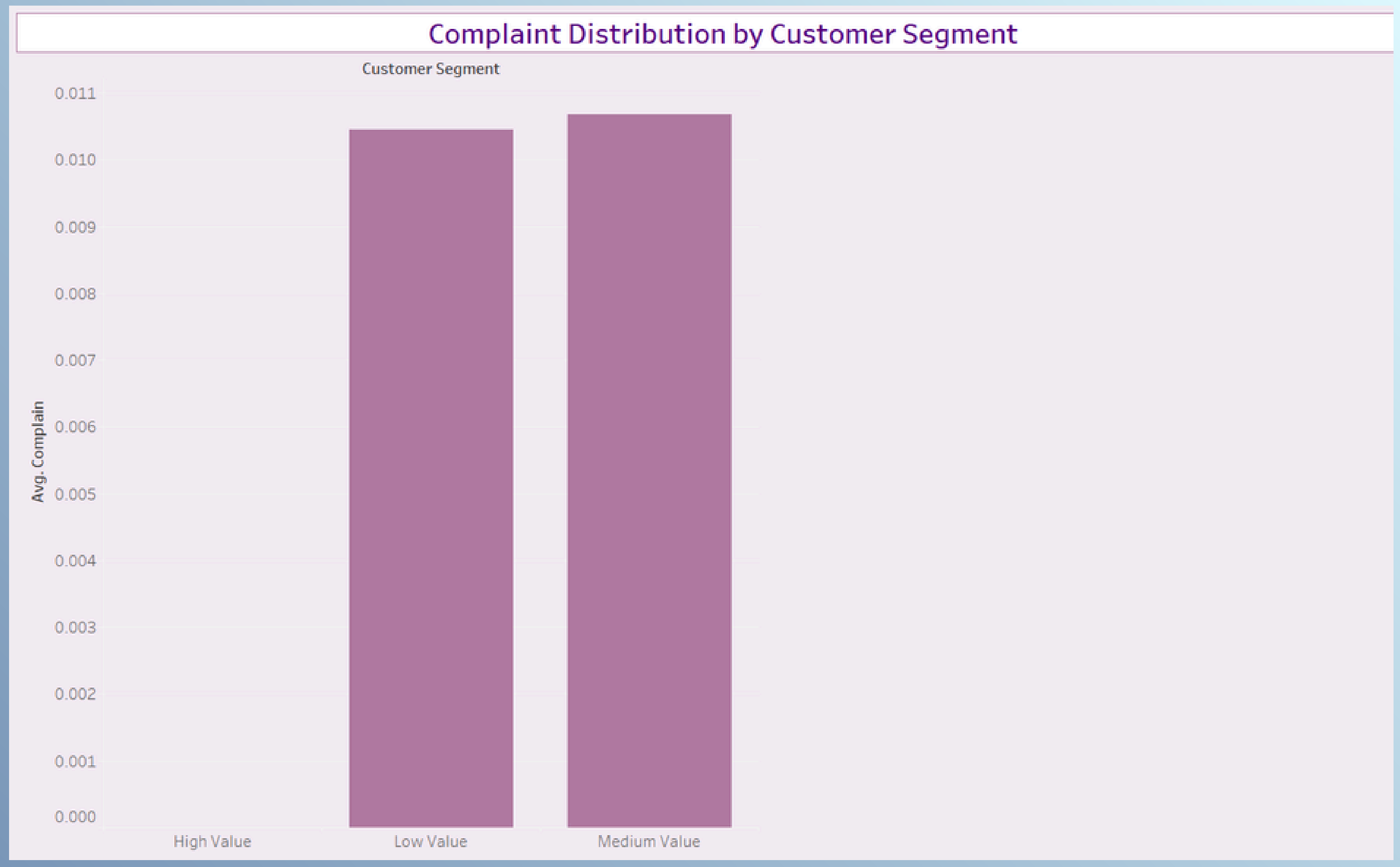
Key Findings:

- Low Value customers recorded the highest number of non-responses.
- Response behavior is relatively balanced across segments.
- High Value and Medium Value customers demonstrate meaningful campaign engagement.

Business Insight:

The overall response rate of 50.38% indicates effective campaign performance. Personalized marketing campaigns can further improve conversion rates among low-response customer groups.

Complaint Distribution by Segment



Complaint rates were analyzed across customer segments to evaluate customer satisfaction.

Key Findings:

- Complaint rates remain extremely low across all customer groups.
- Medium Value customers show slightly higher complaint levels.
- High Value customers reported almost no complaints.

Business Insight:

The overall complaint rate of less than 1% suggests strong customer satisfaction and effective customer relationship management.

Hypothesis Testing

```
t_stat, p_value = ttest_ind(high_income, low_income)

print("T-Statistic:", t_stat)
print("P-Value:", p_value)

if p_value < 0.05:
    print("Reject Null Hypothesis")
    print("Higher-income customers spend significantly more.")
else:
    print("Fail to Reject Null Hypothesis")
```

[12] ✓ 0.0s

```
T-Statistic: 52.441826902856825
P-Value: 0.0
Reject Null Hypothesis
Higher-income customers spend significantly more.
```

Null Hypothesis (H_0):

Customer income has no significant effect on total spending.

Alternative Hypothesis (H_1):

Customer income significantly affects total spending.

Results:

- Median Income = 51,381.5
- T-Statistic = 52.4418
- P-Value = 0.0000

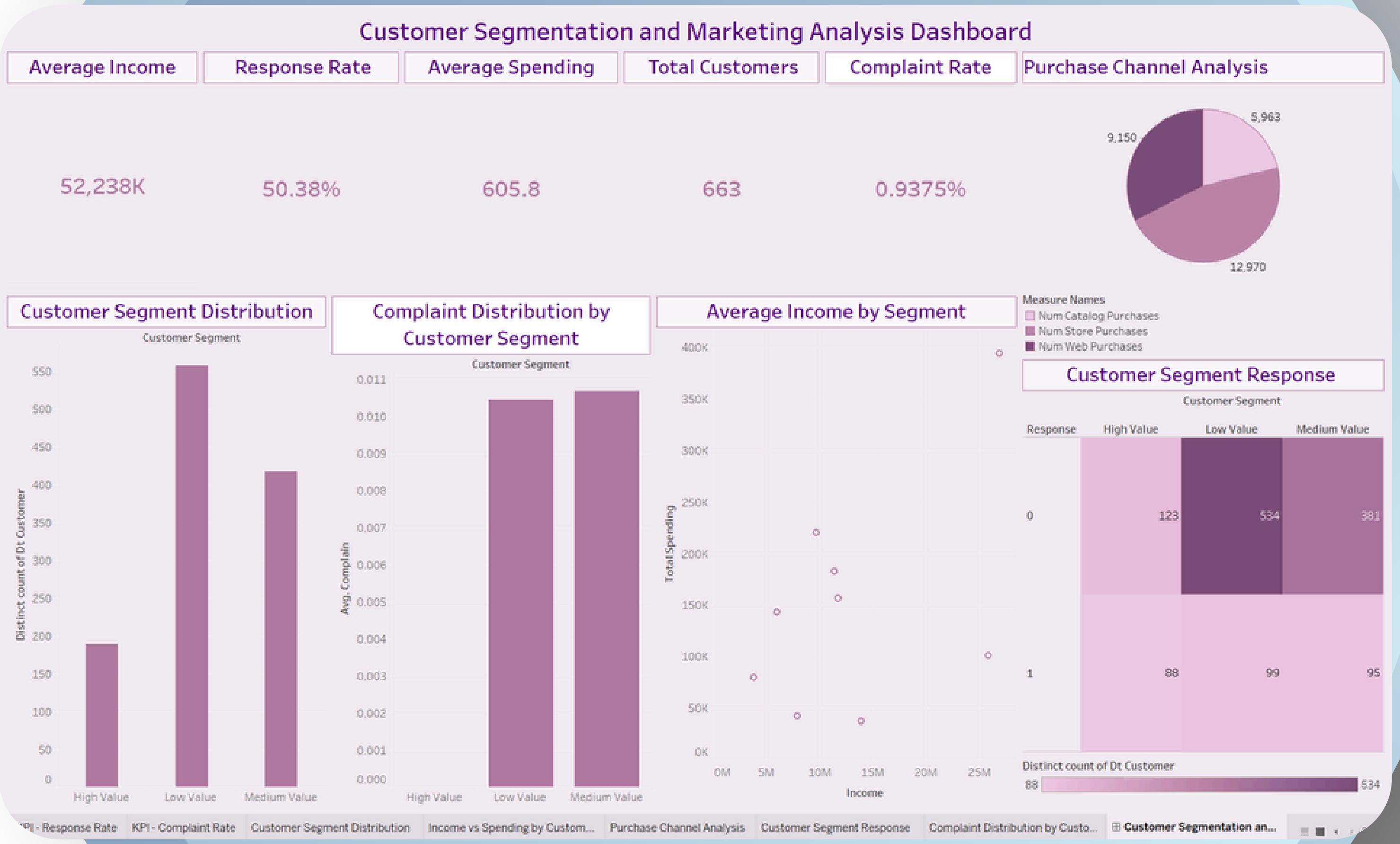
Since the p-value is less than 0.05, **the null hypothesis was rejected**. The analysis confirms that higher-income customers spend significantly more than lower-income customers.

Conclusion

This project leveraged customer segmentation and marketing analytics to uncover key patterns in customer behavior, spending, and engagement.

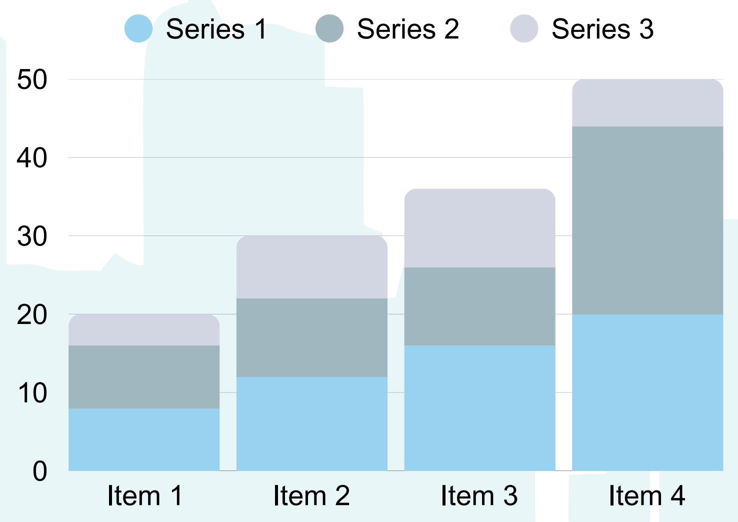
The analysis showed that income significantly influences spending, while customer segmentation and purchase channel analysis provided valuable insights for targeted marketing strategies.

Overall, the dashboard and statistical validation demonstrate how data-driven decision-making can improve customer engagement, marketing effectiveness, and business growth.



Understanding customer behavior through analytics enables smarter marketing decisions and stronger customer relationships.

THANK YOU



Series 1
Series 2
Series 3
50
40
30
20
10
0
Item 1
Item 2
Item 3
Item 4